

Force, mass and acceleration (RP)

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know.

QUESTION	ANSWER	✓
1	Investigating how force affects the acceleration of a trolley: name the independent variable.	Force applied to the trolley ☐
2	Investigating how force affects the acceleration of a trolley: name the dependent variable.	Acceleration of the trolley ☐
3	Investigating how force affects the acceleration of a trolley: name the control variables.	The total mass (trolley and hanging masses) and the surface the trolley travels on ☐
4	Investigating how force affects the acceleration of a trolley: each mass taken off the hanging stack is put on the trolley. Explain why.	To change the force while keeping the total mass the same. ☐
5	The force on a trolley is doubled. State the effect on its acceleration.	It doubles. ☐
6	Investigating how mass affects the acceleration of a trolley: name the independent variable.	Mass of the trolley ☐
7	Investigating how mass affects the acceleration of a trolley: name the dependent variable.	Acceleration of the trolley ☐
8	Investigating how mass affects the acceleration of a trolley: name the control variables.	The force applied and the surface the trolley travels on ☐